Dot Plots

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Introduction

"Dot plot" refers to two distinct chart types that share a name and a geometry but answer different questions. **Cleveland dot plots** show one value per category as a single point, replacing bar charts for ranked comparisons; they were popularised by William Cleveland's 1985 *Elements of Graphing Data* on the grounds that perceiving position is more accurate than perceiving length. **Wilkinson dot plots** show one point per observation, stacked at the relevant x-value, and act as a hand-drawn histogram that exposes individual observations rather than binned counts.

Prerequisites

A working knowledge of `ggplot2`, an idea of when to prefer position to length encoding, and basic familiarity with histograms and bar charts.

Theory

Cleveland dot plots (`geom_point()` with category on y and value on x) eliminate the visual noise of bar lengths and make small differences across many categories easy to compare. A faint horizontal segment from zero to the point — `geom_segment()` — preserves the bar-like sense of "distance from origin" without committing the visual heaviness of a filled bar.

Wilkinson dot plots (`geom_dotplot()`) bin observations along one axis and stack equal-sized dots in each bin. The `method = "histodot"` argument anchors bins to the data range; `method = "dotdensity"` adapts bin centres to local density. They are most useful for small samples ($n < 100$) where every observation should be visible; for larger samples, jittered strip plots or swarm plots scale better.

Assumptions

For Cleveland dot plots: one value per category (or a clearly defined summary). For Wilkinson dot plots: the binwidth makes sense for the data range — too small and dots stack into spikes; too large and they merge into bars.

R Implementation

```r
library(ggplot2)

# Cleveland dot plot: mean mpg by car class
mpg_summary <- aggregate(hwy ~ class, mpg, mean)
ggplot(mpg_summary, aes(hwy, reorder(class, hwy))) +
  geom_segment(aes(x = 0, xend = hwy, yend = class), colour = "grey80") +
  geom_point(size = 3, colour = "#2A9D8F") +
  labs(y = NULL) +
  theme_minimal()

# Wilkinson dot plot: distribution of a small sample
ggplot(mtcars, aes(mpg)) +
```

```r
  geom_dotplot(method = "histodot", binwidth = 1, fill = "#2A9D8F") +
  theme_minimal()

# Grouped dot plot
ggplot(iris, aes(Species, Sepal.Length)) +
  geom_dotplot(binaxis = "y", stackdir = "center", binwidth = 0.1,
               fill = "#2A9D8F") +
  theme_minimal()
```

Output & Results

Three views: a Cleveland dot plot ranking car classes by mean highway mpg with faint baseline segments, a Wilkinson dot plot of the 32 `mtcars` mpg values stacked as a histogram, and a grouped Wilkinson plot showing sepal-length distributions for three iris species side by side.

Interpretation

A reporting sentence (figure caption): "Cleveland dot plot of mean highway fuel economy (mpg) by class for 234 vehicles; categories are ordered by descending mean. Wilkinson dot plot of mpg values for 32 vehicles, with each dot representing a single car." When showing summary points, always describe which summary (mean, median, count) the dot represents.

Practical Tips

- Order Cleveland-dot categories by value with `forcats::fct_reorder()` or `reorder()`; alphabetic orderings rarely communicate ranking.
- Combine Cleveland dot plots with horizontal error bars (`geom_errorbarh()` or `geom_segment()` for confidence intervals) when uncertainty is important.
- Choose Wilkinson dot binwidth to match the data resolution; for integer data, set binwidth equal to the unit step.
- For grouped Wilkinson plots, set `stackdir = "center"` for symmetric stacks or `"up"` for one-sided; `binaxis = "y"` rotates the dotplot for vertical y-axis layouts.
- Dot plots in presentation slides work better than bar charts at small sizes; the bar's length advantage becomes a length-perception headache when the bar is only a few pixels tall.
- Above a few hundred observations per category, switch from Wilkinson dot plots to strip plots, swarm plots (`ggbeeswarm`), or violin plots; stacked dots no longer scale.

R Packages Used

`ggplot2` for `geom_dotplot()`, `geom_point()`, and `geom_segment()`; `forcats` for category reordering; `ggbeeswarm` for swarm-style dot plots that scale to larger samples; `ggdist` for slabinterval geoms that combine the dot-plot and density idioms.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- ggplot2 grammar and multi-panel layouts